Aquifers

Problem 1: Solutions

a) **Specific storage due to water compressibility (S_s^w)**

Using the formula: $S_s^w = \rho_w g n \beta_w$

$$S_s^w = (1000\ \mathrm{kg/m^3}) \times (9.81\ \mathrm{m/s^2}) \times (0.20) \times (4.8 \times 10^{-10}\ \mathrm{m^2/N})$$

$$S_s^w = 9.4 \times 10^{-7}\ \mathrm{m^{-1}}$$

Explanation: This represents the storage capacity due to water expansion when pressure decreases. The small value reflects the low compressibility of water.

b) **Specific storage due to matrix compressibility (S_s^M)**

Using the formula: $S_s^M = \rho_w g \beta_p$

For the range of matrix compressibility values:

Lower bound: $S_s^M = (1000) \times (9.81) \times (5.2 \times 10^{-9}) = 5.1 \times 10^{-5}\ \mathrm{m^{-1}}$

Upper bound: $S_s^M = (1000) \times (9.81) \times (1.0 \times 10^{-8}) = 9.81 \times 10^{-5}\ \mathrm{m^{-1}}$

Range: $S_s^M = 5.1 \times 10^{-5}$ to $9.81 \times 10^{-5}\ \mathrm{m^{-1}}$

Explanation: This represents storage from aquifer compression when water pressure decreases. The matrix compressibility dominates over water compressibility.

c) **Total specific storage and storativity**

Total specific storage: $S_s = S_s^w + S_s^M$

$$S_s = 9.4 \times 10^{-7} + (5.1 \times 10^{-5} \text{ to } 9.81 \times 10^{-5})$$

$$S_s = 5.2 \times 10^{-5} \text{ to } 9.9 \times 10^{-5}\ \mathrm{m^{-1}}$$

Storativity: $S = S_s \times b$

$$S = (5.2 \times 10^{-5} \text{ to } 9.9 \times 10^{-5}) \times 100 \text{ m}$$

$S = 5.2 \times 10^{-3}$ to 9.9×10^{-3} (dimensionless)

Explanation: Storativity represents the volume of water released per unit area per unit head change. The range reflects uncertainty in matrix compressibility.

d) **Volume of water released**

Using: $V = S \times \Delta h \times A$

$$V = (5.2 \times 10^{-3} \text{ to } 9.9 \times 10^{-3}) \times 100 \text{ m} \times 1 \times 10^{9} \text{ m}^2$$

$$V = 5.2 \times 10^{8} \text{ to } 9.9 \times 10^{8} \text{ m}^3$$

Explanation: This represents 520 to 990 million cubic meters of water released from storage due to the 100 m head drop.

e) **Component analysis**

Matrix compressibility dominates: The matrix compressibility term (5.1×10^{-5} to 9.81×10^{-5} m^{-1}) is approximately **50-100 times larger** than the water compressibility term (9.4×10^{-7} m^{-1}).

Physical meaning: In confined aquifers, most water released from storage comes from **aquifer compression** rather than water expansion. When pumping reduces water pressure, the weight of overlying rocks compresses the aquifer matrix, squeezing water out of the pore spaces. This is why confined aquifers have relatively low storativity values compared to unconfined aquifers.

Problem 2: Solutions

a) **Total volume of the soil sample**

$$V_T = \frac{\text{Total mass}}{\rho_b} = \frac{85 \text{ g} + 20 \text{ g}}{1.65 \text{ g/cm}^3} = \frac{105 \text{ g}}{1.65 \text{ g/cm}^3} = 63.6 \text{ cm}^3$$

Explanation: The total mass includes both the drained sample (85 g) and the water that drained out (20 g).

b) **Volume of water retained**

$$V_r = \frac{\text{Mass of retained water}}{\rho_w} = \frac{85\text{ g} - 80\text{ g}}{1.0\text{ g/cm}^3} = \frac{5\text{ g}}{1.0\text{ g/cm}^3} = 5.0\text{ cm}^3$$

Explanation: Retained water is the difference between the drained sample weight and the oven-dried weight.

c) **Volume of water drained**

$$V_d = \frac{\text{Mass of drained water}}{\rho_w} = \frac{20\text{ g}}{1.0\text{ g/cm}^3} = 20.0\text{ cm}^3$$

Explanation: This is the water that drained by gravity during the experiment.

d) **Aquifer properties**

Specific retention: $S_r = \frac{V_r}{V_T} = \frac{5.0\text{ cm}^3}{63.6\text{ cm}^3} = 0.079 = 7.9\%$

Specific yield: $S_y = \frac{V_d}{V_T} = \frac{20.0\text{ cm}^3}{63.6\text{ cm}^3} = 0.314 = 31.4\%$

Porosity: $n = \frac{V_r+V_d}{V_T} = \frac{5.0+20.0}{63.6} = \frac{25.0}{63.6} = 0.393 = 39.3\%$

e) **Verification of relationship**

$n = S_y + S_r = 31.4\% + 7.9\% = 39.3\%$ ✓

Explanation: This confirms that porosity equals the sum of specific yield and specific retention, as expected.

f) **Material classification**

Classification: This material has **high porosity** (39.3%) and **high specific yield** (31.4%), characteristic of **coarse-grained sediments** like sand or sandy gravel.

Aquifer potential: This material would make an **excellent aquifer** because:

- High specific yield means most water can be extracted by pumping
- Low specific retention (7.9%) means minimal water is held against gravity
- High porosity provides substantial storage capacity

Comparison to Table 4.1: These values are similar to sand (porosity 25%, specific yield 22%) but higher, suggesting this might be a well-sorted, coarse sand or sandy gravel.

Problem 3: Solutions

a) **Basin-Fill Aquifers**

Formation process: Basin-fill aquifers form in structural basins between mountain ranges. In the Basin and Range province, **tectonic extension** creates north-south trending mountain ranges separated by valleys. **Erosion** of the mountains provides sediment that fills the valleys, creating thick accumulations of alluvial material.

Geometry: Basin-fill aquifers are characterized by:

- **Thick accumulations** (hundreds to thousands of meters)
- **Elongated, narrow bodies** following the valley structure
- **Complex layering** with interbedded coarse and fine materials

Mountain-aquifer relationship: The mountains act as **sediment sources** and **recharge areas**. Snowmelt and runoff from mountain slopes provide both water and sediment to the basin-fill aquifers.

b) **Fluvial Aquifers**

Braided vs meandering systems:

- **Braided rivers:** Multiple, shifting channels with high sediment loads create **continuous, well-connected** aquifer systems. The high-energy environment produces **coarse-grained deposits** with high permeability.
- **Meandering rivers:** Single, sinuous channels with lower sediment loads create **discontinuous, lens-shaped** aquifers. The lower-energy environment produces **finer-grained deposits** with lower permeability.

Braided river advantage: Braided systems create more productive aquifers because:

- **Lateral continuity** of coarse deposits
- **Higher hydraulic conductivity** due to coarse grain size
- **Better connectivity** between different parts of the aquifer

Sedimentological characteristics: Good fluvial aquifers are characterized by:

- **Coarse grain size** (sand and gravel)
- **Good sorting** (uniform grain size)
- **High porosity** and permeability
- **Lateral continuity** of permeable units

c) **Comparison**

Hydraulic properties:

- **Basin-fill aquifers:** Variable hydraulic conductivity due to complex layering; can be very high in coarse units but limited by fine-grained interbeds
- **Fluvial aquifers:** Generally high hydraulic conductivity in braided systems; more variable in meandering systems

Heterogeneity:

- **Basin-fill aquifers:** High heterogeneity due to complex depositional history and multiple sediment sources
- **Fluvial aquifers:** Lower heterogeneity in braided systems; higher in meandering systems due to channel migration

Development suitability:

- **Basin-fill aquifers:** Better for **large-scale development** due to thick accumulations and large storage capacity
- **Fluvial aquifers:** Better for **local development** with high yields but limited extent

Sustainability:

- **Depositional environment controls sustainability** through:
  - **Recharge rates** (mountain sources vs river recharge)
  - **Storage capacity** (thick basins vs thin fluvial deposits)
  - **Water quality** (mountain sources vs river contamination)

Problem 4: Solutions

a) **High Plains Aquifer System**

Geographic extent: The largest aquifer in the United States, underlying ~450,000 km² across parts of Colorado, Kansas, Nebraska, New Mexico, Oklahoma, South Dakota, Texas, and Wyoming.

Geological composition: Primarily the **Ogallala Group** (late Miocene, 5-16 million years old), consisting of unconsolidated gravel, sand, silt, and clay deposited by eastward-flowing braided streams.

Depositional history: The **ancestral Rocky Mountains** provided sediment source. Eastward-flowing streams deposited alluvial sediments in accommodation space created by tectonic subsidence. Local salt dissolution and faulting in southwestern Kansas created additional subsidence and thicker deposits.

Current usage: In 2000, ~30% of groundwater pumped for irrigation in the US came from this aquifer. **Overpumping** is a significant sustainability issue.

b) **Central Valley Aquifer System**

Surface water system: Unique system where most rivers flow **westward from the Sierra Nevada** into the Central Valley. The Sacramento River flows south, and the San Joaquin River flows northward to the Delta and San Francisco Bay.

Alluvial processes: Westward-flowing rivers from the Sierra Nevada (Kings, Kaweah, Kern Rivers) formed alluvial fans. The **glaciated portions** of the Sierra Nevada (Kings River fan) produced coarse sediments, while **non-glaciated watersheds** (Fresno River, Chowchilla fans) produced finer-grained deposits.

Variability:

- **Coarse-grained deposits** near mountain fronts (alluvial fans)
- **Finer-grained deposits** in valley bottoms and along the Coast Range
- **Permeable units** associated with San Joaquin River channels
- **Hydraulic conductivities** range from 0.07 to 1000 m/day

c) **Indo-Gangetic Basin (IGB) Aquifer**

Geographic extent: Vast flat plain drained by the Indus and Ganges Rivers and their tributaries, extending across Pakistan, India, and Bangladesh.

Sedimentological complexity: The cross-sections show:

- **Pleistocene alluvium** (older, yellow sediments) forming the main aquifer body

- **Holocene deposits** (younger, orange sediments) as incised channels within the Pleistocene
- **Complex layering** with interchannel deposits, fluvial deposits, and piedmont deposits
- **Active subsidence** in the Bengal Basin creating additional accommodation space

World importance: This aquifer is considered the world's most important because:

- **Vast extent** covering multiple countries
- **High population density** dependent on groundwater
- **Agricultural importance** for food security
- **Complex hydrogeology** requiring careful management

Challenges: Over-exploitation, contamination, and transboundary management issues.

d) **Comparative Analysis**

Depositional environments:

- **High Plains:** Continental interior, braided stream deposits from mountain erosion
- **Central Valley:** Intermontane basin with alluvial fans from mountain sources
- **IGB:** Large river system with complex fluvial and deltaic deposits

Factors for importance:

- **High Plains:** Large extent and storage capacity
- **Central Valley:** High agricultural productivity and water demand
- **IGB:** Critical for food security and population support

Sustainability influences:

- **Geological history** determines recharge rates and storage capacity
- **Current tectonic activity** affects subsidence and water quality
- **Climate change** impacts recharge and demand patterns
- **Human activities** (pumping, contamination) affect long-term viability

Key insight: All three systems face sustainability challenges due to over-exploitation, but their geological characteristics provide different opportunities for management and conservation strategies.